

DATTA SAI V V N

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Summary

Data Engineer and **AI/ML Engineer** with almost 4 years of experience in **production ML systems**, data engineering, and predictive analytics using Python, PyTorch, TensorFlow and SQL. Skilled in building end-to-end solutions across Deep Learning, NLP, and time-series forecasting with expertise in MLOps and model deployment. Combines strong statistical foundations with hands-on experience delivering scalable, data-driven solutions.

Education

University of Illinois Chicago, Chicago, USA

Aug 2024 – May 2026

Master of Science in Computer Science

Relevant Coursework: Data Science, AI Engineering, Network Communication, System Engineering, Machine Learning

Skills

Programming Languages & Tools: Python, SQL, R, Scala, Bash, Git, Jupyter, VS Code, Antigravity, Linux

Machine Learning: Scikit-learn, XGBoost, LightGBM, Random Forest, Regression, Classification, Clustering

DL & NLP: PyTorch, TensorFlow, Keras, BERT, GPT, Transformers, LSTM, CNNs, spaCy, Hugging Face

Data Engineering: Spark, Airflow, Kafka, Hadoop, Hive, ETL Pipelines, Data Warehousing, PostgreSQL, MySQL

MLOps & Cloud: MLflow, Docker, Kubernetes CI/CD, Netlify, FastAPI, AWS, Azure ML, Model Deployment

Analytics: Power BI, Excel, Matplotlib, Seaborn, Plotly, A/B Testing, Statistical Analysis, KPI Dashboards

Experience

Data Engineer

Mar 2025 – Present

University of Illinois Chicago, USA

- Built automated ETL pipelines for StarREZ housing data to generate occupancy reports for 1,000+ units.
- Developed SQL queries and Python scripts to analyze occupancy patterns, vacancy rates, and turnover metrics.
- Created Power BI dashboards visualizing trends and forecasts, reducing manual reporting time by 40%.

Data Engineer

Aug 2022 – Aug 2024

Alstom Transportation, Bangalore, India

- Built ML and OCR pipeline to extract and classify requirements from customer documents using Python and NLP.
- Engineered features and trained models with Scikit-learn and PyTorch to improve accuracy and reliability.
- Deployed FastAPI-based inference service on AWS for real-time requirement classification and allocation.
- Automated reporting and traceability dashboards in Power BI to monitor model performance and compliance.

Research Intern – Artificial Intelligence /Machine Learning

Jun 2022 – Aug 2022

Indian Institute of Technology Ropar, India

- Preprocessed and analyzed research datasets using Python, Pandas, and NumPy for ML pipeline development.
- Implemented supervised and unsupervised learning models including XGBoost, SVMs, and clustering algorithms.
- Validated models with cross-validation and visualized results using Matplotlib and Seaborn for reproducibility.

Data Science Intern

Jan 2022 – Jun 2022

Ernst & Young, Chennai, India

- Built predictive models using Regression and Decision Tree algorithms to forecast IT service response time.
- Automated data collection and preprocessing pipelines with Python, Pandas, and SQL for analytics reporting.
- Deployed ML workflow on ServiceNow data to identify process bottlenecks and improve efficiency metrics.
- Visualized model outcomes and KPIs using Power BI dashboards for data-driven decision-making by leadership.

Projects

Real-Time Deepfake Detection | *Python, PyTorch, ResNeXt-50, RNN, MTCNN, Docker, AWS, Streamlit*

- Built deepfake detection pipeline using ResNeXt-50 CNN with MTCNN facial extraction and temporal sampling.
- Trained ensemble model on 140K videos achieving 96% accuracy with data augmentation and transfer learning.
- Deployed production system with Streamlit and Docker, enabling real-time inference at 30+ FPS with GPU.

Predictive Maintenance with LSTM Time-Series Forecasting | *Python, LSTM, TensorFlow, Spark, Airflow*

- Engineered predictive solution combining survival analysis with LSTM networks, processing 50K+ records.
- Built automated ETL pipeline with Spark and Airflow for time-series feature engineering and data transformation.
- Achieved 85% accuracy predicting failures 90 days ahead, reducing unplanned downtime by 20% for operations.